THE BETTER POLICY PROJECT

AI-POWERED LEARNING

Next Generation of Policy Frameworks and Modeling

WEEK 2 · JUNE 15-19, 2026

DSGE Models for

Monetary and Fiscal Policy

GIMF & Global Scenarios

The IMF's full-scale tool for multi-country fiscal-monetary scenarios — including spillovers and policy coordination.

Chulalongkorn University · Bangkok, Thailand

GIMF

What is GIMF?

The IMF's working tool for multi-country fiscal and monetary policy questions — used for real scenarios, not just academic exercises

STRUCTURE

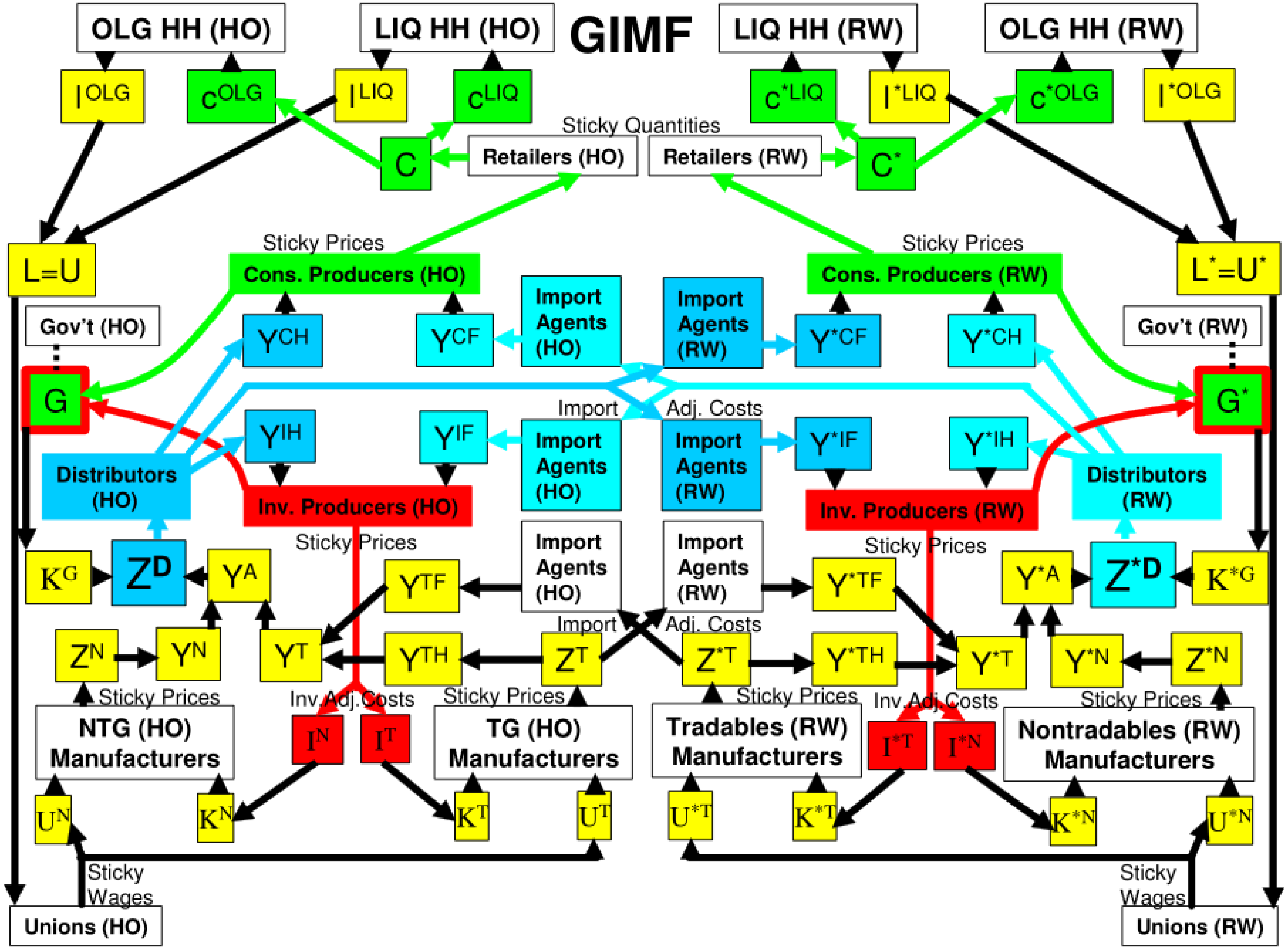
Multi-country DSGE

- 20+ region world: US, EA, Japan, EM, rest of world
- OLG households + liquidity-constrained households
- Tradables + non-tradables firms in each region
- Government with detailed fiscal block
- Bilateral trade and capital flows

PURPOSE

Built for fiscal-monetary scenarios

- Fiscal multipliers for different instruments
- Long-run effects of debt on potential output
- International spillovers from fiscal shocks
- Coordination scenarios (e.g. G20 stimulus)
- Monetary-fiscal policy interaction at the ZLB


GIMF
OLG HH (HO)
LIQ HH (HO)
LIQ HH (RW)
OLG HH (RW)
I^{OLG}
C^{OLG}
I^{LIQ}
C^{LIQ}
C^{*LIQ}
I^{*LIQ}
C^{*OLG}
I^{*OLG}
Sticky Quantities
C
Retailers (HO)
Retailers (RW)
C^*
L=U
$L^*=U^*$
Sticky Prices
Cons. Producers (HO)
Sticky Prices
Cons. Producers (RW)
Gov't (HO)
Gov't (RW)
Y^{CH}
Y^{CF}
Import Agents (HO)
Import Agents (RW)
Y^{*CF}
Y^{*CH}
G
G^*
Import
Adj. Costs
Y^{IH}
Y^{IF}
Import Agents (HO)
Import Agents (RW)
Y^{*IF}
Y^{*IH}
Distributors (HO)
Inv. Producers (HO)
Inv. Producers (RW)
Distributors (RW)
Sticky Prices
Import Agents (HO)
Import Agents (RW)
Sticky Prices
K^G
Z^D
Y^A
Y^{TF}
Y^{*TF}
Y^{*A}
Z^{*D}
K^{*G}
Import
Adj. Costs
Z^N
Y^N
Y^T
Y^{TH}
Z^T
Z^{*T}
Y^{*TH}
Y^{*T}
Y^{*N}
Z^{*N}
Sticky Prices
Inv. Adj. Costs
Sticky Prices
Sticky Prices
Inv. Adj. Costs
Sticky Prices
NTG (HO) Manufacturers
I^N
I^T
TG (HO) Manufacturers
Tradables (RW) Manufacturers
I^{*T}
I^{*N}
Nontradables (RW) Manufacturers
U^N
K^N
K^T
U^T
U^{*T}
K^{*T}
K^{*N}
U^{*N}
Sticky Wages
Sticky Wages
Unions (HO)
Unions (RW)

GIMF

Four reasons Ricardian equivalence breaks in GIMF

Each break enriches the model and increases the realism of fiscal effects

01

Finite horizons

OLG households with mortality probability p — debt repayment partly falls on future generations not yet alive.

02

Liquidity constraints

Share λ of households cannot borrow against future income — they consume current income, MPC ≈ 1 on transfers.

03

Distortionary taxes

Labor and consumption taxes affect prices, marginal incentives, and supply. Lump-sum equivalence fails.

04

Productive G

Government investment adds to public capital, which raises potential output. The level of G has real long-run effects.

GIMF

Agent structure

Three household types, two firm types, one government — per region

HOUSEHOLDS	**OLG savers**	Forward-looking. Mortality p. Hold financial wealth. Consumption depends on $(W+H)\cdot(\theta+p)$.
HOUSEHOLDS	**Liquidity-constrained**	Cannot smooth — consume current disposable income. Share λ of population. MPC ≈ 1.
FIRMS	**Tradables sector**	Faces world prices. Compete in international markets. Calvo-style price setting.
FIRMS	**Non-tradables sector**	Faces only domestic demand. Pricing power in domestic market. Sticky prices.
GOVERNMENT	**Fiscal authority**	Uses 7 fiscal instruments. Subject to fiscal rule (structural surplus). Issues debt, collects taxes.
MONETARY	**Central bank**	Taylor rule responding to inflation and output gap. Smoothing parameter ρ. Monetary shock ε_m.

GIMF

The fiscal rule

GIMF disciplines fiscal policy with a structural surplus rule — twin objectives

STRUCTURAL SURPLUS RULE

$$def_t = d^* + \varphi_b (b_{t-1} - b^*) + \varphi_y (y_t - y^*)$$

deficit responds to deviation of debt from target (debt stabilization) and output from potential (cyclical stabilization)

PURPOSE I

Debt stabilization

- Target the interest-inclusive deficit ratio
- Implies a target debt-to-GDP ratio b*
- Mean reversion is slow: 1/(g+n) ≈ 0.99
- Long-run debt anchored, but not rigidly

PURPOSE II

Business cycle stabilization

- Save excess tax revenue during booms
- Run deficit during recessions
- Rules-based representation of automatic stabilizers
- Higher ϕ_y → more countercyclical stance

GIMF

The seven fiscal instruments

GIMF treats fiscal policy as a portfolio — different instruments, different transmission

1
SPENDING
Government Investment

2
SPENDING
Government Consumption

3
SPENDING
General Transfers

4
SPENDING
Targeted Transfers

5
TAX
Labor Income Tax

6
TAX
Consumption Tax

7
TAX
Corporate Income Tax

GIMF

What determines the size of the multiplier?

There is no such thing as 'the' fiscal multiplier — eight factors drive the outcome

1

Monetary accommodation

Larger when policy doesn't react

2

Persistence of stimulus

Larger for longer-lived shocks

3

Direct vs indirect

Larger for G than for transfers/taxes

4

Hand-to-mouth share

Larger when λ is high

5

Targeting of transfers

Larger when reaching LC households

6

Economic openness

Smaller in more open economies

7

Nominal rigidities

Larger when prices/wages stickier

8

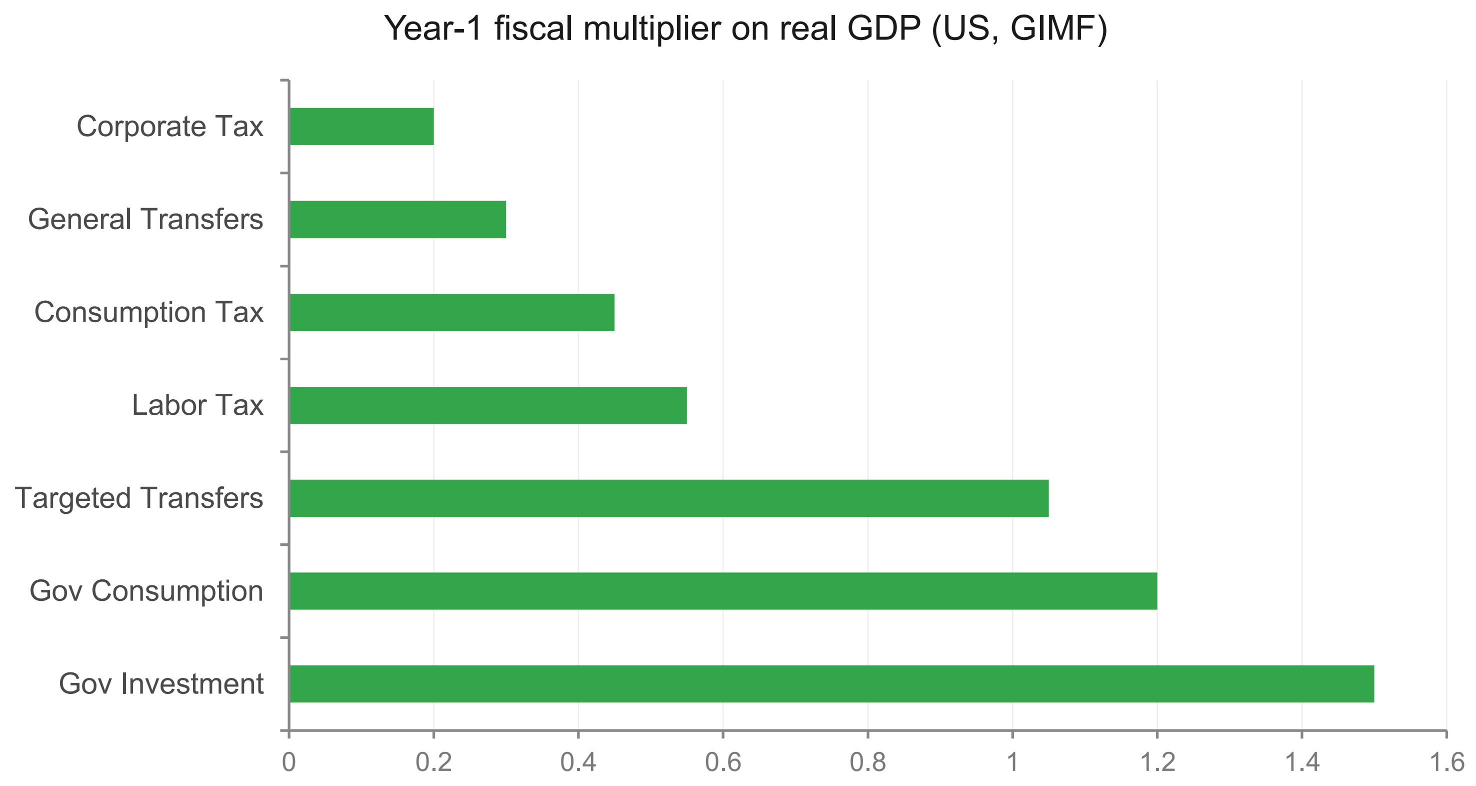
Initial slack

Larger when output is below potential

GIMF

Temporary multipliers — direct vs indirect

Why government investment moves output much more than corporate tax cuts



THE PATTERN

Direct demand

Government investment and consumption hit aggregate demand directly. Multipliers > 1.

Targeting matters

Targeted transfers reach hand-to-mouth households. General transfers leak to savers.

Indirect channels

Tax cuts and corporate measures must work via private behavior — much smaller multipliers.

GIMF

Permanent fiscal changes

Different from temporary stimulus — long-run crowding-out becomes the dominant story

LONG-RUN COSTS

Negative effects on potential output

- Higher debt → higher real rates → less private capital
- Higher distortionary taxes → lower labor supply
- Lower government investment → lower public capital
- Composition matters: financing through tax cuts to investment vs labor income vs consumption
- Effects compound over decades

THE COMPOSITION QUESTION

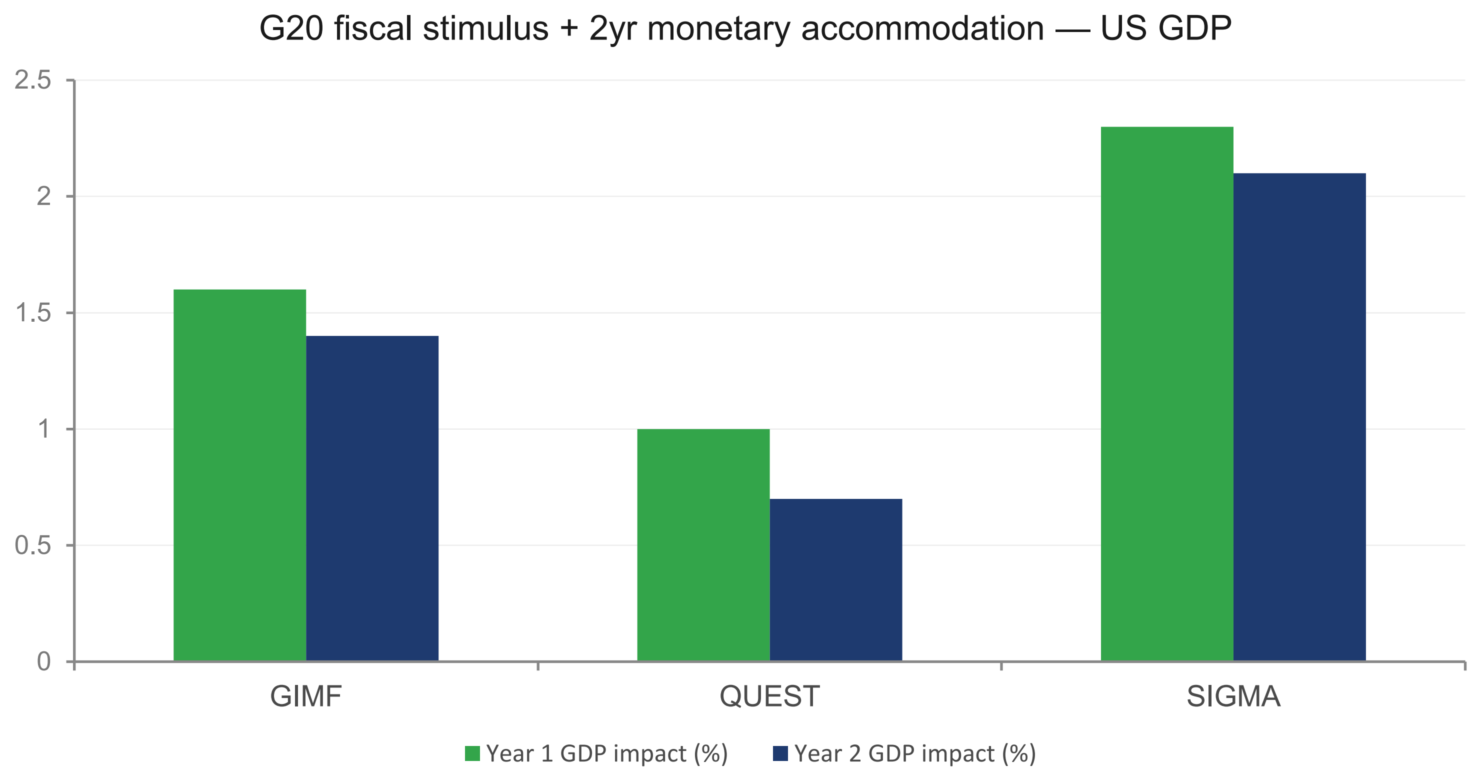
Which spending? Which tax?

- Cutting investment to fund interest = lasting damage to potential output
- Raising labor tax = persistent labor supply distortion
- Raising consumption tax = less distortionary, but still real cost
- Raising corporate tax = significant capital-deepening loss
- These are quantitatively different — GIMF lets you compare

GIMF

G20 fiscal stimulus — model comparison

Same scenario, three structural models, three answers — and a useful range for policymakers



WHY THE DIFFERENCES?

SIGMA (largest)

Larger international spillovers, more inflation response.

GIMF (middle)

Balanced inflation persistence, medium spillovers.

QUEST (smallest)

Less inflation movement, more crowding-out under monetary accommodation.

TOPIC 4 · GIMF

End-of-week capstone

Pick a real fiscal-monetary question your institution faces — and use the right DSGE model to answer it

01

Pick a real question

Something your Board would actually want answered. 'How much fiscal room do we have?' or 'What if the US slows hard?' or 'Is our transmission still working?'

02

Match model to question

Sustainability questions → OLG. Spillover questions → GIMF. Transmission questions → NK. Let the question pick the model.

03

Run the analysis

Calibrate to your country. Apply the relevant shock or experiment. Read what the model is telling you — and where it's silent.

04

Brief the Board

Two pages: the question, what the model said, where you'd be cautious, what it means for the next round.

By the end of the week: a complete scenario-based policy note grounded in real model output.

Key Takeaways

1

A real working tool

GIMF is what the IMF actually uses for major fiscal-monetary scenarios. Multi-country, several agent types, seven fiscal instruments — built for the questions a country sees in real life.

2

There is no single multiplier

Eight things change the answer — accommodation, persistence, instrument, hand-to-mouth share, openness, rigidities, slack, targeting. Anyone quoting one number is hiding something.

3

Composition matters

Government investment moves output most. Targeted transfers reach the people who'll spend. Tax cuts work indirectly and slowly.

4

Short-run vs long-run

Temporary stimulus boosts demand. Permanent debt costs you on the supply side — through capital, taxes, and potential output.

References

Source paper and the core methodological literature

- Anderson, D., B. Hunt, M. Kortelainen, M. Kumhof, D. Laxton, D. Muir, S. Mursula, and S. Snudden. 2013, "Getting to Know GIMF – The Simulation Properties of the Global Integrated Monetary and Fiscal Model," IMF Working Paper No. 13/55.
- Kumhof, M., D. Laxton, D. Muir and S. Mursula, 2010, "The Global Integrated Monetary Fiscal Model (GIMF)---Theoretical Structure", IMF Working Paper 10/34 (February 2010)
- Fiscal Policy References on our The Better Policy Project Website: https://www.thebetterpolicyproject.org/research-1

Great Things are Coming.

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